Riphah School of Computing & Innovation

Updated FYP Proposal Evaluation Rubrics

Category: IoT & Embedded Systems Project

These rubrics have been explicitly designed to align with the structural requirements of 'FYP Framework v1.0' and 'Template03 - Project Proposal & Plan -v1.1'. Key additions include evaluations for Bill of Materials (BOM), Hardware-Software Interface, Environmental Tolerance, and Edge/Cloud Processing.

Domain A: Problem Identification & Background Study (40 Marks)

Criteria & Max Marks	Unsatisfactory (0–1)	Developing (2–3)	Proficient / Excellent (4–5)	Marks
1. Executive Summary & Problem Formulation (10 Marks)	Missing executive summary. Problem statement is vague, undefined, or lacks real-world physical/automation context.	Summary is present but lacks core hardware/software highlights. Problem is stated but lacks clear contextual constraints.	Clear executive summary covering core functionality, hardware stack, and main deliverable. Precise, well-formulated problem statement addressing a clear physical or automation need.	(10 Marks)
2. Proposed Solution & Functional Scope (10 Marks)	No clear functional requirements or scope. Edge vs. cloud operations are vaguely described.	Lists basic features but lacks clear traceability to user/system needs. Misses explicit non-functional requirements (NFRs) like latency or power.	Explicitly details Functional Requirements, specific edge-logic vs cloud processing split, and clear NFRs (e.g., battery life, response time, data frequency).	(10 Marks)
3. Existing System & Literature Analysis (10 Marks)	Minimal background study. Fails to review current manual processes or existing IoT solutions.	Summarizes existing solutions but lacks rigorous competitor analysis, hardware comparison, or clear gap identification.	Comprehensive critical synthesis of state-of-the-art competitors, existing systems, or manual methods, explicitly establishing the technical, hardware, and functional gap.	(10 Marks)
4. Target Environment & Constraints (5 Marks)	Target deployment environment is unidentified. No operating constraints stated.	Environment is mentioned but lacks analysis of physical constraints (e.g., connectivity issues, weather).	Clearly defined physical deployment conditions, network availability assumptions, and power source availability with clear mitigation strategies.	(5 Marks)

5. Project Plan (WBS, Gantt & Roles Matrix) (5 Marks)	No WBS, Roles Matrix, or Gantt chart provided. Hardware procurement timeline missing.	Basic timeline provided but missing explicit hardware bench/field testing phases or unclear team responsibilities.	Detailed WBS including procurement, bench testing, and field pilot. Clear Roles & Responsibility Matrix, and structured Gantt chart with task assignments.	(5 Marks)

Domain B: Technical Design & Methodology (35 Marks)

Criteria & Max Marks	Unsatisfactory (0–1)	Developing (2–3)	Proficient / Excellent (4–5)	Marks
7. System Architecture & Communication Protocols (8 Marks)	No architecture diagram provided; communication protocols are unspecified.	Conceptual block diagram present, but hardware-software interface, specific protocols, or payload structure are missing/unclear.	Clear multi-tier architectural diagram detailing physical nodes, gateways, and cloud. Well-justified communication protocol (MQTT, BLE, LoRa, etc.) and explicit data payload structure.	(8 Marks)
8. Bill of Materials (BOM) & Sourcing (7 Marks)	Hardware components are generic (e.g., 'sensor', 'microcontroller'). No budget estimation.	Lists some specific components but lacks full BOM details, unit costs, or sourcing viability.	Complete BOM with exact part/model numbers, quantities, accurate estimated unit costs, total hardware budget, and notes on local availability/sourcing.	(7 Marks)
9. Hardware-Software Interface & Edge Logic (5 Marks)	Unclear how hardware interacts with software. No edge processing defined.	Basic explanation of data transmission, but lacks detail on power constraints or edge vs cloud decision-making logic.	Explicit definition of edge logic (what runs on-device vs cloud), precise transmission frequency, and clear power/connectivity constraint management.	(5 Marks)
10. Environmental Tolerance & Failure Handling (8 Marks)	No mention of physical conditions or hardware failure scenarios.	Mentions environmental factors but lacks a concrete testing plan (bench vs field) or automated fallback handling.	Detailed plan for environmental tolerance (e.g., IP ratings, temperature). Explicit testing plan (bench validation before field pilot) and robust failure handling (e.g., local caching on network drop).	(8 Marks)
11. Hardware Novelty & Practical Viability (7 Marks)	Direct clone of a standard tutorial IoT project. No practical value or viability.	Slight modification of standard sensor arrays. Practical impact or real-world deployment viability is limited.	High practical utility, clear feature/hardware enhancements over existing tools, cost-effective design, and strong potential for a successful real-world pilot.	(7 Marks)

Domain C: Oral Defense & Presentation (25 Marks)

Criteria & Max Marks	Unsatisfactory (0–1)	Developing (2–3)	Proficient / Excellent (4–5)	Marks
12. Presentation Organization & Visual Quality (5 Marks)	Unstructured, cluttered slides, unreadable text, missing architectural/circuit diagrams.	Logical structure but visual aids are standard or overly text-dense. Lacks clear BOM visual breakdown.	Exceptionally structured slides with clean layout, high-contrast hardware/network diagrams, and executive-ready aesthetics.	(5 Marks)
13. Technical Communication & Delivery (10 Marks)	Reads directly from slides, poor time management, weak grasp of embedded systems vocabulary.	Clear delivery but relies on notes; adequate time management, unbalanced team speaking roles regarding hardware vs software.	Fluent, engaging delivery, precise technical/hardware terms, perfect time management, and balanced team participation across all domains.	(10 Marks)
14. Q&A Response & Hardware Design Defense (10 Marks)	Unable to answer technical questions; cannot justify component selection or network protocols.	Answers surface-level questions but struggles to justify deep architectural choices, edge/cloud split, or BOM alternatives.	Demonstrates deep mastery; logically defends microcontroller/sensor choices, protocol efficiency, scope boundaries, and fail-safes against panel scrutiny.	(10 Marks)